Exp1

Exp1.1

Editor - C:\Users\abhiz\Desktop\5g-6g-communication\exp1.m

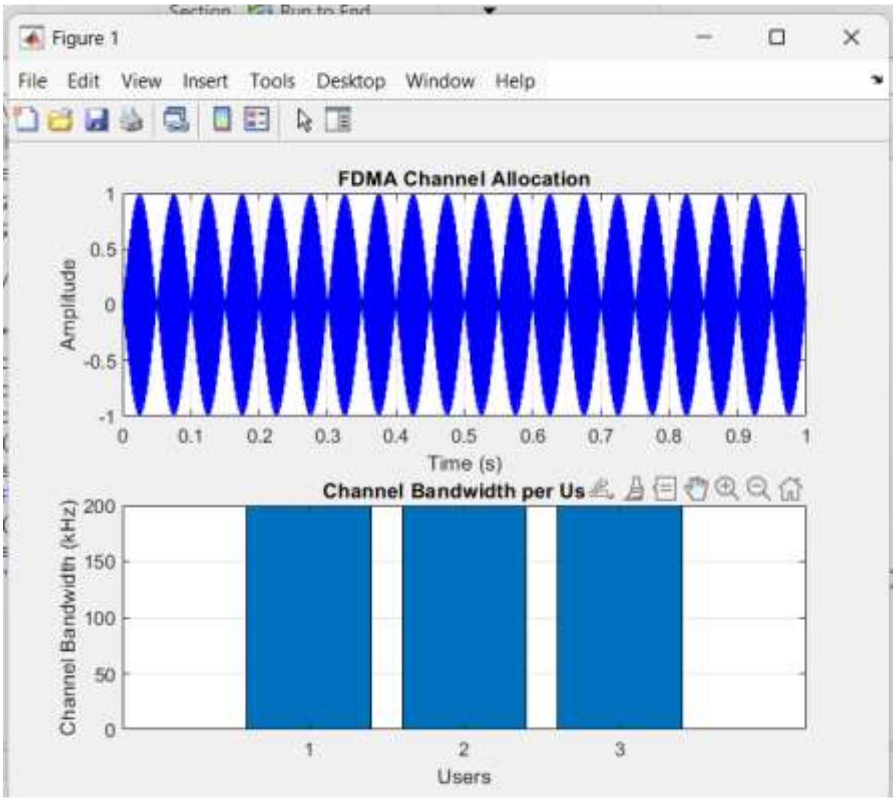
exp1.m | exp1_obj_2.m | exp1_obj_3.m | exp2_obj_1.m | exp2_obj_2.m | exp2_obj_3.m

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;

CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on
xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

Command Window

```
User Wise Slot Allocation
    User     Status(1=Allocated,0=Blocked)
    ____     _____________________________

     1                  1
     2                  1
     3                  1
     4                  1
     5                  1
```

OUTPUT



EXP-1.2

Editor - C:\Users\abhiz\Desktop\5g-6g-communication\exp1_obj_2.m

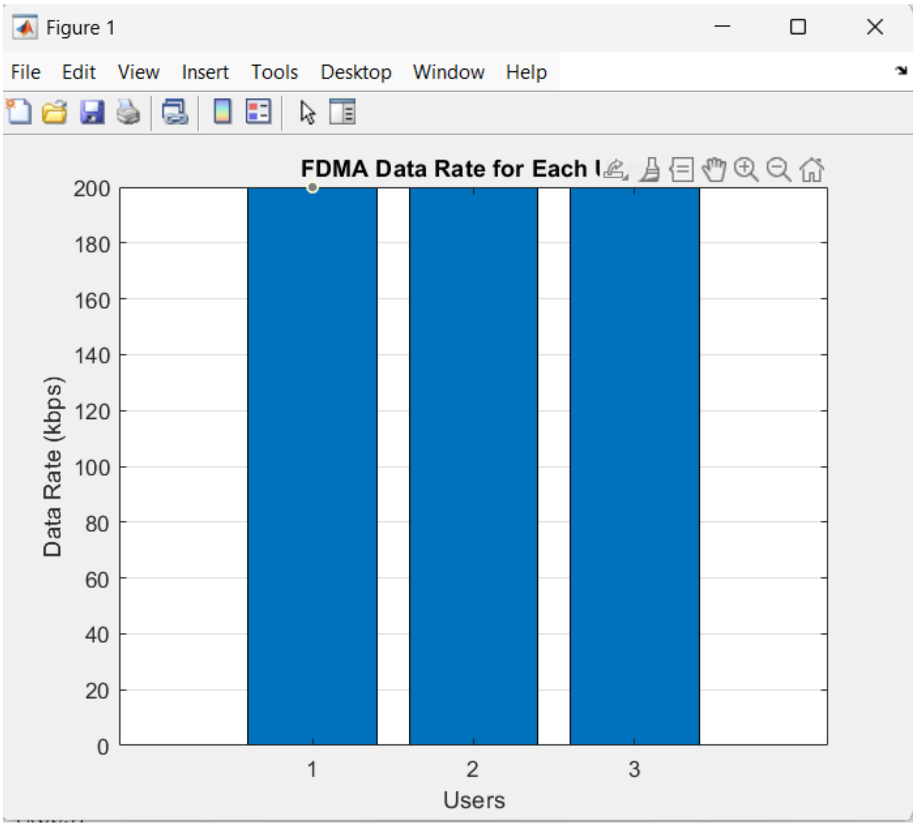
exp1.m | exp1_obj_2.m | exp1_obj_3.m | exp2_obj_1.m | exp2_obj_2.m | exp2_obj_3.m

```
clc; clear; close all;
BW = 200; % Channel Bandwidth (kHz)
Users = 3;
M = 2; % BPSK Modulation
DataRate = BW*log2(M); % Data Rate (kbps)
Rate = DataRate*ones(1,Users);
disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on
```

Command Window

```
 User Data Rate (kbps)
     1   200
     2   200
     3   200

>>
```

OUTPUT



EXP-1.3

```
clc; clear; close all;
Ps = 1; % Signal Power (W)
Pn = [0.1 0.05 0.01]; % Noise Power (W)
Users = 1:3;
SNR = 10*log10(Ps./Pn); % SNR in dB
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```

Command Window

```
User Noise Power(W) SNR(dB)
    1.0000    0.1000   10.0000
    2.0000    0.0500   13.0103
    3.0000    0.0100   20.0000

>>
```

OUTPUT